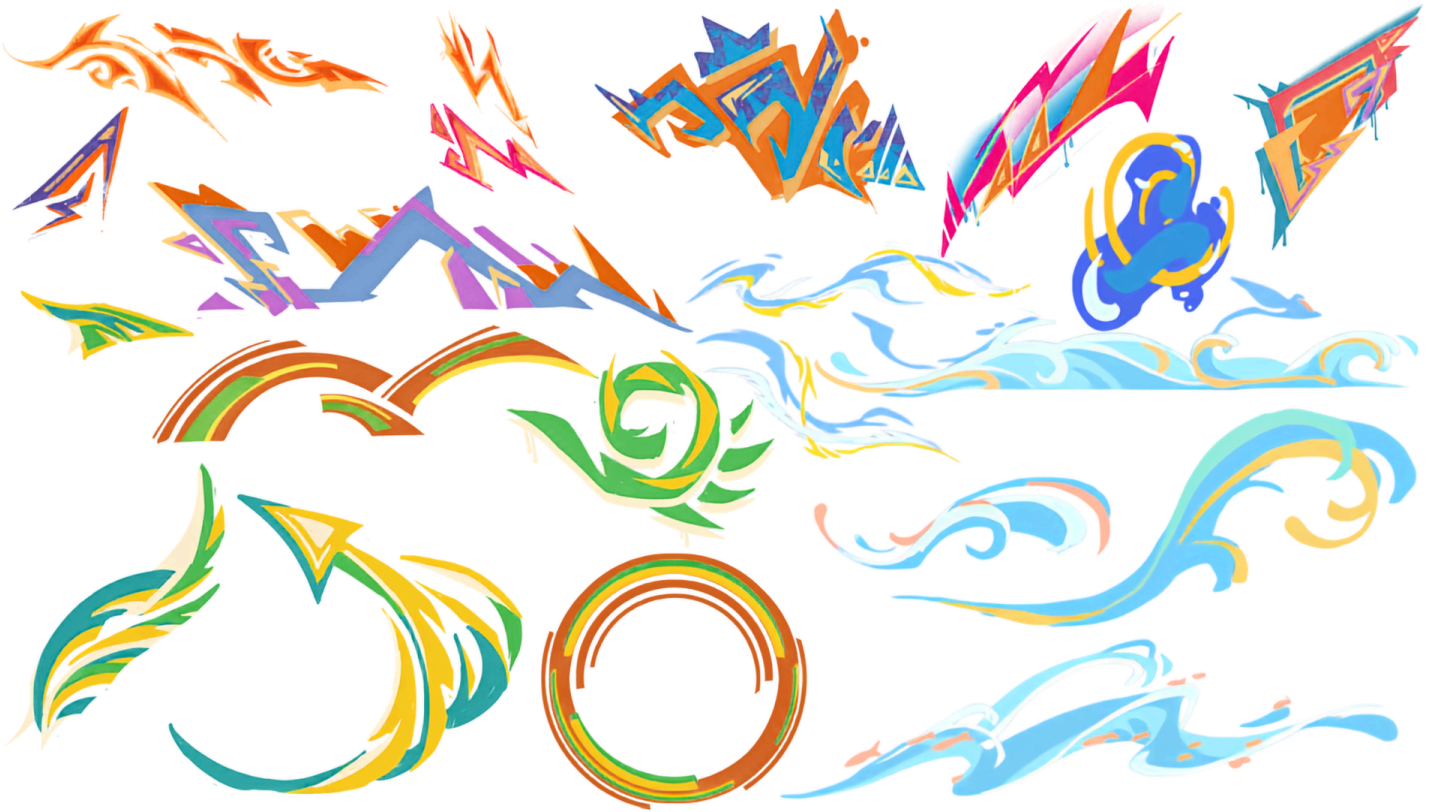


GeckApp: Invitation

By Kha Bui

References

- **Visual & Art Style (Natlan Theme):** Inspired by *Genshin Impact's Natlan region* — features vibrant graffiti art, tribal patterns, and warm neon hues (burnt orange, cyan, glowing red) for UI elements, frames, and background motifs.



- **Minigame 1 - Art Team (Color Match):** Referenced from casual color-picker tools like *Figma/Photoshop Color Picker* and games like *Color Match 3D* — players slide RGB bars to match a randomized target color.
- **Minigame 2 - Media Team (Snap the Moment):** Referenced from *Pokémon Snap* — players tap the screen to photograph a mascot at the exact moment it enters the viewfinder frame and strikes a pose.
- **Minigame 3 - Dev Team (Bitmask Switch):** Referenced from *Unity Layer Mask System* and *Cyberpunk 2077's Breach Protocol* — a minimal UI-based puzzle where players toggle 0/1 binary bits to match a target bitmask.

Target Platform

- **Target Platform:** Mobile Devices (iOS & Android smartphones via Mobile Web Browsers for instant QR Code access).
- **Display Orientation:** Mobile Portrait Mode (9:16 ratio) designed for convenient one-handed play at event booths.

Development Tools



Game Engine: Unity (Version 2022.3 LTS or 2023.2 LTS) for cross-platform WebGL build and lightweight 2D performance.

UI & Art Tools: Aseprite (UI/UX wireframes and layout design) / Adobe Illustrator (Graffiti, textures, and asset exports).

Project Management & Code: GitHub (version control & collaboration) & Visual Studio Code (scripting and repository management).

Game Overview

Concept: A fast-paced mobile WebGL game designed to introduce the club's structure and departments through 3 quick minigames, rewarding players with tokens to spin a Gacha system for club information and merch.

Story: Players roleplay as a new student visiting the club booth, completing tasks for the Dev, Art, and Media teams to earn Gacha tokens and discover everything about the club.

Gameplay Loop: Scan QR Code → Play Department Minigames (Dev, Art, Media) → Earn Gacha Tokens → Spin Gacha to collect Club Info cards or win physical Merch at the booth.

Objectives:

- *Short-term:* Complete each minigame to collect maximum Gacha tokens.
- *Long-term:* Spin the Gacha to unlock all club information cards (high chance) and win real physical club merch (rare chance).

Gameplay Mechanics/Systems

Gacha & Token System (Core Mechanics):

- **Token Earning:** Players earn 1 Gacha Token upon successfully completing any department minigame.
- **Gacha Spin:** Spend 1 Token to spin the Gacha machine.
- **Drop Probability:** High chance (95%) for **Club Info Cards** (department lore, activities, culture); Low chance (5%) for **Physical Merch Vouchers** (stickers, keychains, badges) redeemable at the booth.
- **Collection Book:** A digital album tracking unlocked Club Info Cards

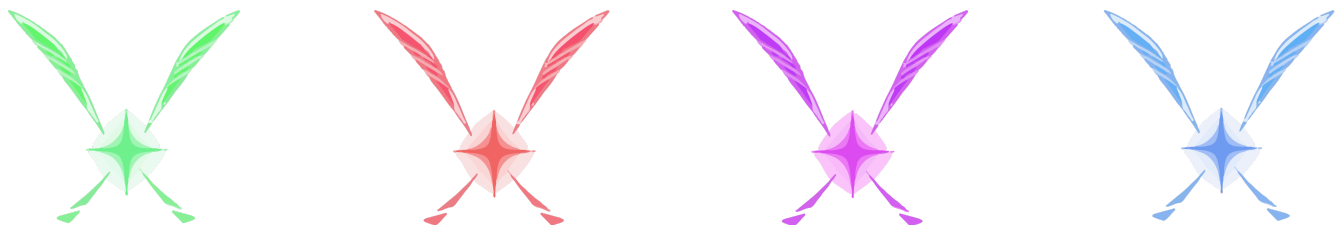
Minigame 1 - Art Department ("Color Match"):

- **Mechanic:** The system presents 3 randomized target colors simultaneously. Players adjust 5 UI sliders (**Red, Green, Blue, White, Black**) to mix and match all 3 target colors.
- **Win Condition:** Achieve a overall color accuracy for all 3 colors within a 90-second time limit to earn 1 Token(Limit: 1 Token max).
- **Controls:** Touch and drag UI sliders (100% Mobile Touch).



Minigame 2 - Media Department ("Snap the Moment"):

- **Mechanic:** The camera viewfinder frame remains stationary in the center while the club mascot moves across the screen, briefly stopping to strike a pose.
- **Win Condition:** Tap the shutter button at the precise moment the mascot poses inside the viewfinder frame. Capture 10 successful photos within 60 seconds to earn 1 Token(Limit: 1 Token max).
- **Controls:** Single-tap anywhere on the screen or on the shutter UI button.



Minigame 3 - Dev Department ("OOP Object Inspector"):

- **Mechanic:** The system displays a target Object name generated from a 3-boolean matrix (canJump, canSwim, canAttack). Players toggle 3 attribute switches (ON/OFF) to construct the requested Object:

FLY	SWIM	ATTACK	OBJECT / ENEMY
FALSE	FALSE	FALSE	Villager_NPC

TRUE	FALSE	FALSE	Rabbit_Animal
FALSE	TRUE	FALSE	Goldfish_Pet
FALSE	FALSE	TRUE	Statue_Turret
TRUE	TRUE	FALSE	Frog_Amphibian
TRUE	FALSE	TRUE	Goblin_Warrior
FALSE	TRUE	TRUE	Shark_Monster
TRUE	TRUE	TRUE	Super_Boss

Win Condition: Correctly toggle attributes and tap INSTITUTE() to match the target Object 3 times in a row within 15 seconds to earn 1 Token (Limit: 1 Token max).

Controls: Single-tap on 3 attribute toggle switches (ON và OFF)

- ON - Star / Green Glow
- OFF - Blue Out

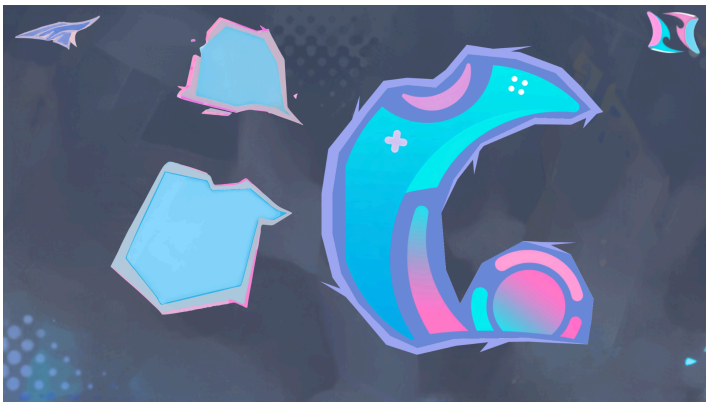


UI (reference)

Main Screen



Minigame UI - Gacha



Minigame UI - Dev Team



Minigame UI - Art Team



Minigame UI - Media Team



Team Breakdown & Responsibilities

Dev Team (2 Members - Technical & Gameplay Systems):

- **Dev 1 (Core System & Backend Logic):**
 - Implement Gacha wheel mechanics, Token tracking logic, and local storage/collection book saving.
 - Handle build optimization for smooth mobile browser performance via QR code access.
- **Dev 2 (Minigames Logic):**
 - Program the math/logic for all 3 minigames: RGBWB slider color accuracy, camera viewfinder hit-timing, and the 3-boolean OOP Object matrix.
 - Integrate audio triggers, UI event listeners, and victory/defeat state transitions.

Optional - Art Team (1 Member - UI & Visual Design):

- **Art Lead (UI/UX & Asset Creation):**
 - Design all UI buttons, frames, HUD elements, and interactive components across minigames using **GeckApp**.
 - Export optimized graphic assets (Main screen UI, Gacha card frames, Mascot sprites, and custom ON/OFF toggle buttons).
 - Maintain visual consistency and layout alignment for portrait mobile screens.

